NOT MEASUREMENT SENSITIVE

MIL-PRF-62321D
20 March 1997
SUPERSEDING
MIL-T-62321C
15 August 1989

PERFORMANCE SPECIFICATION

TRUCKS, TANK: POTABLE WATER, 1000 AND 2200 GALLON, DIESEL AND GASOLINE ENGINE DRIVEN, 19 000 AND 34 500 POUNDS GVW, 4x2, 4x4, AND 6x4, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, two-rear-wheel drive, and four-wheel, four-wheel drive, potable water tank trucks with a 1000 gallon capacity, having a minimum GVW of 19 000 pounds for normal operating conditions; and diesel engine driven, six-wheel, four-wheel-drive, 2200 gallon capacity potable water tank trucks, having a minimum GVW of 34 500 pounds for normal operating conditions. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2320

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1.2 Classification. The tank trucks are one of the models and sizes shown in table I, as specified (see 6.2):

TABLE I. Vehicle classification.

Model (wheels and wheels driving)	Size (gallons)	GVW (pounds)
4x2	1000	19 000
4x4	1000	19 000
6x4	2200	34 500

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-2128 - Reflector Set, Highway Warning, Triangular.
A-A-50271 - Plate, Identification.
A-A-52550 - Pintle Assembly, Towing, Manual Release, 18 000, 40 000 and 100 000 Lbs Capacity.
A-A-52557 - Fuel Oil, Diesel; for Posts, Camps and Stations.
A-A-52592 - Pipe Fittings: One or More Ends Grooved.
A-A-52598 - Couplings, Clamps, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe.

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STANDARDS

DEPARTMENT OF DEFENSE

MS75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.
MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Code of Federal Regulations	- Title 49, Part 178.
Code of Federal Regulations	- Title 49, Parts 393 and 399.
Code of Federal Regulations	- Title 49, Part 571.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations	- Title 40, Part 86.

FOOD AND DRUG ADMINISTRATION (FDA)

Code of Federal Regulations	- Title 21, Part 177.

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations	- Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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DRAWINGS

11677011	- First Aid Kit.

(Copies of the above drawing are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)

Standards Manual

(Application for copies of the European Tyre and Rim Technical Organisation publications should be addressed to the European Tyre and Rim Technical Organisation, 32 Avenue Brugmann-B, 1060 Brussels, Belgium.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A240	- Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD Adopted).
SAE J537	- Storage Batteries (DoD Adopted).
SAE J551/1	- Performance Levels and Methods of Measurement of Electromagnetic Compatibility of Vehicles and Devices (60 Hz to 18 GHz).

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SAE J551/2	- Test Limits and Methods of Measurement of Radio Disturbance Characteristics of Vehicles, Motorboats, and Spark-Ignited Engine-Driven Devices.
SAE J560	- Seven Conductor Electrical Connector for Truck-Trailer Jumper Cable (DoD Adopted).
SAE J588	- Turn Signal Lamps for Use on Motor Vehicles Less Than 2032 mm in Overall Width (DoD Adopted).
SAE J589	- Turn Signal Switch (DoD Adopted).
SAE J682	- Rear Wheel Splash and Stone Throw Protection (DoD Adopted).
SAE J688	- Truck Ability Prediction Procedure (DoD Adopted).
SAE J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DoD Adopted).
SAE J706	- Rating of Winches (DoD Adopted).
SAE J844	- Nonmetallic Air Brake System Tubing (DoD Adopted).
SAE J1067	- Seven Conductor Jacketed Cable for Truck Trailer Connections (DoD Adopted).
SAE J1349	- Engine Power Test Code - Spark Ignition and Compression Ignition-Net Power Rating (DoD Adopted).
SAE J2014	- Pneumatic Tires for Military Tactical Wheeled Vehicles.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

THE TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 175 Montrose W Avenue, Suite 150, Copley, OH 44321.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.12, the vehicle, components, assemblies and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.12, no removal, substitution, or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with best commercial practices. Primer paint thickness shall be at least 2 mils. Finish paint thickness shall be at least 3 mils. Total paint finish shall be at least 5 mils thick. The exterior color and the identification markings shall be as specified in the contract for the military service identified by the procuring activity (see 6.2). When specified (see 6.2), concealed markings shall be furnished. The tank may be painted to match the color of the truck chassis.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with best commercial practices. If tropical rustproofing is required, it shall be as specified in the contract (see 6.2).

3.1.1.3 Drain plugs. Drain plugs installed in manual transmissions, transfer case and rear axles shall be of the permanent magnet type.

3.1.1.4 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins, or the chassis manufacturer's standard single, center mounted eye or pin shall be mounted on the front of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

3.1.1.5 Drive hubs. When specified for 4x4 model vehicles (see 6.2), manually operated drive hubs, for disengaging the front wheels from the power train, shall be provided when available, except for vehicles equipped with interaxle compensating devices between front and rear axles (see 3.4.5.1.2).

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3.1.1.6 Wheel splash and stone throw protection. All tilt cabs shall have rubber mud flaps to the rear of the front wheels. A two-inch wide (minimum) backing strip of corrosion-resistant material shall be used to prevent mudflap fastening hardware from damaging the mudflaps. All splash shield and mud flap installations shall conform to the rear wheel splash and stone throw protection provisions as specified in SAE J682.

3.1.1.7 Trailer towing package. When specified (see 6.2), a trailer towing package consisting of a pintle, safety chain attachment devices, a seven-pin lighting receptacle in accordance with SAE J560, and when specified (see 3.4.11.2), a trailer brake control system, and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to AA52550-1 for 4x2 and 4x4 model vehicles, and AA52550-2 for 6x4 model vehicles. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 1800 pounds and a horizontal drawbar load of not less than 18 000 pounds for 4x2 and 4x4 models; and a vertical tongue load of not less than 4000 pounds and a horizontal drawbar load of not less than 40 000 pounds for 6x4 models directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than four inches forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. For 4x2 and 4x4 model vehicles, each attachment device shall provide an ultimate strength of not less than 18 000 pounds, and the devices shall be capable of accommodating a standard grab hook (2.19 inches wide, 0.73 inch thick and 0.48 inch throat width) for a 3/8-inch chain. For 6x4 model vehicles, each attachment device shall provide an ultimate strength of not less than 40 000 pounds, and the devices shall be capable of accommodating a standard grab hook (4.56 inches wide, 1.19 inches thick, 0.78 inch throat width) for a 5/8-inch chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color-coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.8 Trailer lighting cable. When specified (see 6.2), a seven-pin, trailer lighting cable conforming to SAE J1067 shall be furnished. The cable shall be coiled and shall be not less than 110 inches long when fully extended. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

3.1.1.9 Brushguard. Model 4x4 vehicles shall be equipped with a radiator and headlamp brushguard. When the headlamps are recessed in and protected by the front bumper, a headlamp brushguard is not required.

3.1.1.10 Identification and instruction plates. A power takeoff caution notice and body operating controls, caution, warning and instruction plates shall be provided. Plates shall conform to composition A, class 1 or 2, or composition C of A-A-50271.

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3.1.1.11 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

3.1.1.12 4x4 conversion. The chassis manufacturer's standard 4x2 truck chassis may be modified to provide all wheel drive conforming to the requirements specified herein if:

a. The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer.
b. The chassis manufacturer's front axle before conversion has the same load rating as the conversion axle to be installed.
c. Components used in the all wheel drive conversion are of current production.
d. Components used in the all wheel drive conversion are approved for the conversion application by the component manufacturers.
e. The converted vehicle conforms to Federal Motor Vehicle Safety Standard, Title 49, Part 571, Safety Standard No. 121.
f. Replacement headlights, if required to be added to meet the height requirements of Federal Motor Vehicle Safety Standard, Title 49, Part 571, Safety Standard No. 108, are equivalent in mounting, protection, and range and precision of adjustment to the chassis manufacturer's original standard headlights.
g. Unused headlight cavities are covered in a neat workmanlike manner, treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis cab manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.1.1.2.
h. Complete installation drawings are made available to procurement after award of the contract.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

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3.2.2 Air pollution control. The vehicle shall comply with Title 40, Part 86, EPA regulations governing control of air pollution from new motor vehicles and new motor vehicle engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. Noise levels shall be less than 85 decibel (A-Weighted Scale) (dB(A)) at the operator and occasionally occupied positions during typical operations and environments. This shall include those locations where personnel are operating or monitoring pumping operations. If noise levels exceed 85 dB(A), then noise hazard warning decals that are visible to all personnel exposed to the hazard shall be provided stating that hearing protection is required.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories and equipment; the body; and a full complement of fuel, lubricants and coolant (see 4.1.2).

3.2.5 Gross vehicle weight. Gross vehicle weight shall consist of curb weight, operator weight (determined at 175 pounds), and a payload to provide not less than the specified GVW. Vehicle with a four-door crew cab shall have GVW determined with 525 pounds of payload in the rear seat. The rated payload capacity shall be not less than the weight determined from the rated liquid volume, using 8.3 pounds per gallon (see 4.1.2).

3.2.6 Weight distribution. The distribution of GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the truck body load area (see 4.1.2).

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. The minimum GVW rating shall be 19 000 and 34 500 pounds, respectively. In addition, the actual GVW rating of the vehicle furnished shall exceed the minimum GVW rating specified (19 000 or 34 500) if required to provide for a payload weight equal to 8.3 pounds per gallon times the size of the tank body in gallons (see table I).

3.2.8 Dimensions. The vehicle shall have a rear-of-cab to rear axle dimension of not less than 83 inches for 4x2 and 4x4 model vehicles, and a cab-to-trunnion dimension of not less than 120 inches for 6x4 model vehicles. The cab-to-trunnion dimension may be reduced by not more than two inches when the vehicle is furnished with a tilt type cab (see 3.4.12). The overall width of the vehicle, exclusive of tires and safety related items such as mirrors, lights and reflectors, shall be not more than 96 inches. The width-over-tires shall be not more than 100 inches.

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3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.2.10 Prohibited materials. Asbestos, cadmium, and radioactive material shall not be used in any part of this vehicle. Radioactive material is defined by Code of Federal Regulations, Title 10, Parts 30 and 40 and is defined as other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries.

3.3 Performance. Performance requirements for 4x4 model vehicles shall be met with front wheel drive disengaged and with front wheel driving hubs unlocked, except that vehicles equipped with interaxle compensating devices shall meet performance with front wheel drive engaged (see 4.1.3).

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the vehicle loaded to specified GVW (see 4.1.3).

3.3.1.1 High speed gradeability. The 4x2 and 4x4 model vehicles shall ascend a continuous grade of 1.5 percent at 50 miles per hour (mph), and 6x4 model vehicles shall ascend a continuous grade of 0.5 percent at 50 mph. Gradeability requirements shall be met with the main transmission in direct drive and, when a multispeed axle is furnished, with the axle in high speed range. Gradeability shall be determined in accordance with SAE J688 (see 4.1.3).

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be determined with engine operating at not less than 35 percent of recommended governed speed, and shall provide not more than the vehicle speed specified in table II for corresponding number of available forward speeds (see 4.1.3).

TABLE II. Vehicle low speed requirement.

Number of forward speeds	Maximum low speed (mph)		
	4x2 model	4x4 model	6x4 model
4 or 5	4.5	--	--
8	3.5	3	3.0
9 or 10	3	2.5	3.0
13	--	--	2.5

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 55 mph for 4x4 model vehicles and 58 mph for all other vehicles (see 4.1.3).

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3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 20 percent grade. The service brakes shall stop the vehicle, loaded to the specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation, Title 49, section 393.52 (see 4.1.3).

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower used in performance predictions shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to A-A-52557. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-8 fuel under normal conditions (see 6.3). A power loss when operating on JP-8 is acceptable.

3.4.1.2 Gasoline engine. When specified for 4x2 and 4x4 model vehicles (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed of not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Engine net horsepower used in performance predictions shall be determined in accordance with SAE J1349. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's recommended operating speed.

3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10 000 feet above sea level or in an ambient air temperature of not less than

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125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two quart capacity. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level (see 4.1.3).

3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof slave receptacle for receiving external power and grounding vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation, Title 49, section 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures down to minus 60°F without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose, excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

a. A coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed.
b. An immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in oil pan through any convenient opening.
c. A battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperature as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

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d. A fuel warmer or pre-heater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulation, Title 49, sections 393.27 through 393.31 and 393.33.

3.4.2.1 Starting system (diesel). For diesel engine driven vehicles, a 12-volt or if available a 24-volt direct current (dc) starting system with 12-volt lighting system, and with not less than a 60 ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Ignition system (gasoline). For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished. An alternator of not less than 40 amperes rated capacity and which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished. When a converter type 24-volt system is provided for a trailer receptacle (see 3.4.2.6), an alternator of not less than 100 amperes capacity which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished.

3.4.2.3 Lighting. All vehicle lights, reflectors and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulation, Title 49, sections 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails unless recessed and fully protected or mounted on vehicle bumpers.

3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have a visible and audible type flash indicator.

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3.4.2.5 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings and the total cold cranking ratings at 0°F, both measured in accordance with SAE J537, shall be not less than as specified in table III. The batteries shall be of the maintenance-free type.

TABLE III. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel (less than 220 gross horsepower)	360	1500
Diesel (220 gross horsepower and over)	630	1800
Gasoline	110	600

3.4.2.6 Auxiliary 24-volt system with trailer receptacle. When specified (see 6.2), an auxiliary 24-volt system with a trailer receptacle assembly shall be furnished. Either a converter type (see 3.4.2.6.1) or an alternator type (see 3.4.2.6.2) system meeting specified requirements shall be furnished. A trailer receptacle, conforming to MS75021-2, with cover assembly shall be provided in an accessible location on the rear end of the vehicle. A twelve-conductor cable, not less than 110 inches long with both ends of cable equipped with connectors conforming to MS75020-1 and MS75020-2, shall be furnished. The cable assembly shall be stowed in the vehicle. The 24-volt, 12 ampere output, service lighting circuit shall be connected through appropriate lighting controls to terminals B, D, E, J and L of MS75021-2.

3.4.2.6.1 Converter type 24-volt system. The 12- to 24-volt converter shall operate from the 12-volt battery(s) (see 3.4.2.5). The output capacity shall be not less than 12 amperes.

3.4.2.6.2 Alternator type 24-volt system. The alternator type 24-volt system shall be separate from the 12-volt vehicle lighting and ignition system and shall include:

a. Nominal 24-volt alternator with not less than 25 amperes rated capacity and capable of providing not less than 7 amperes dc output at normal engine idle speed.
b. Two 12-volt batteries with a combined capacity of at least 40 ampere-hours at a 20-hour rate or one 24-volt battery with at least 20 ampere-hours capacity at a 20-hour rate.
c. Voltage regulating device.
d. An ammeter, mounted on the instrument panel.

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3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551/1 and J551/2. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulation, Title 49, sections 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished.

3.4.3.2 Fuel tank(s). The vehicle shall be equipped with step type fuel tank(s) of not less than 50 gallons total capacity (43 gallons for vehicles destined for California). When more than one tank is furnished for gasoline or diesel engine driven vehicles, means shall be provided to assure equalized fuel levels in both tanks. As an alternative for gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel intake can be provided which includes a means to monitor the level of either tank from a single gage.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation, Title 49, section 393.83. Vertical exhaust mufflers, if furnished and if capable of being reached easily by personnel entering or leaving either side of the cab, shall be provided with a heat shield. Vertical exhaust systems shall be provided with a hinged rain cap.

3.4.5 Gear train.

3.4.5.1 Manual transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be provided with power take-off opening(s) conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

3.4.5.1.1 Speeds. The vehicle shall be provided with not less than five forward speeds for 4x2 model vehicles, not less than eight forward speeds for 4x4 model vehicles, and not less than 9 forward speeds for 6x4 model vehicles. The vehicle may be furnished with an auxiliary transmission.

3.4.5.1.2 Transfer case. A transfer case shall be provided on model 4x4. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between front and rear axles, the transfer case shall provide for driver selection of either two-wheel or four-wheel drive. When furnished, interaxle compensating devices shall provide for positive transfer of power to all driving axles. When a two speed transfer case is furnished, the speedometer shall read accurate speed (mph) with the speed selector in high or low range.

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3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the model vehicle and engine furnished, with the clutch torque capacity exceeding the maximum delivered engine torque. The clutch lining shall be asbestos free.

3.4.5.3 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The transmission shall include a hydraulic torque converter and not less than five forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with an SAE J704 power takeoff opening. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

3.4.5.4 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be installed so as to be readily visible to the seated driver (see 3.1.1.10).

3.4.6 Driveline components. The driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction, divided equally between the driving axles furnished.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the vehicle furnished. The chassis frame rails shall not project beyond the rear end of the body. For 6x4 model vehicles, reinforcements extending at least from the rear of the front suspension rear hanger bracket to the bogie trunnion mounting bracket shall be furnished.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. When the suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure build-up and for replenishing the oil supply shall be provided.

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3.4.9.1 Two-speed axle. When specified for 4x2 or 6x4 model vehicles (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. The gear ratio shall provide performance specified in 3.3.1 through 3.3.1.3.

3.4.9.2 Rear bogie. A rear bogie shall be furnished on 6x4 model vehicles. The rear bogie shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall permit differential action between the two axles, and shall be provided with a manually or automatically controlled lockout assuring equal power to each rear axle. Manual lockout control used shall be located in the truck cab. Axle gear ratios shall provide the performance specified in 3.3.1 through 3.3.1.3.

3.4.9.3 Traction control. When specified (see 6.2), a traction control shall be furnished on the rear axle (either rear axle for model 6x4). The traction control shall actuate automatically to ensure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle (see 4.1.3).

3.4.10 Wheels, rims, tires and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations, for the type and size of tires furnished. Multi-piece rims shall not be furnished. Tire size, tire load range (ply rating), and rim sizes shall be the same for all wheels on each vehicle. When specified (see 6.2), wide base type tires and wheels for the front and rear axles shall be provided in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. Ten-stud disc type wheels shall be furnished. When specified (see 6.2), cast spoke type wheels shall be furnished.

3.4.10.1 Tires. Tires shall conform to the Tire and Rim Association (TRA) or TRA Military Supplement dimension standards for the tire design and size selected. The selected tire shall also have the TRA or TRA Military Supplement load-inflation rating equal to or above the maximum tire position load on the vehicle for the maximum vehicle speed capability. Tires furnished under this specification shall be products conforming to the requirements in table I and the selected requirements of table 2 in SAE J2014. The technical activity for the vehicle system shall determine the mission profile of the vehicle. Per the vehicle's mission profile, the tire technical activity shall determine which tests from table 2 of SAE J2014 are required and should be selected to evaluate potential candidate tires. For procurement or replacement tires, the tire size, load, and percentage of usage for primary, secondary, and off-road mobility shall be included in the solicitation package by the procuring activity (see 6.2). This information shall be provided to the contractor by the vehicle manufacturer prior to tire and/or vehicle qualification.

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3.4.10.2 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.16.1.

3.4.10.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare wheel or rim or a spare tire assembly is furnished on a vehicle without a carrier, it shall be stowed securely on the vehicle for shipment. When a spare tire assembly is specified (see 3.4.10.4), a spare wheel or rim shall be furnished.

3.4.10.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the vehicle.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulation, Title 49, sections 393.40 through 393.43 and 393.45 through 393.52. Model 4x2, 4x4, and 6x4 vehicles shall be provided with full air brakes on all wheels. Brake linings shall be of nonasbestos material (see 4.1.3).

3.4.11.1 Air brakes. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute for 4x2, 4x4, and 6x4 model vehicles.
b. Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between compressor and the last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air control valves.
e. Air pressure gage, visible to the driver.
f. Low air pressure warning, visible and audible.
g. Service brake stop lamp switch.
h. Automatic moisture ejector.
i. For 4x2 and 6x4 models only, automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes.

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3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Trailer brake control system. When specified (see 6.2), a trailer brake control system shall be furnished when a trailer towing package is specified (see 3.1.1.7). The trailer brake control system shall include:

a. Identification of emergency and service lines.
b. Coincident control of trailer brakes with prime mover foot control.
c. Independent hand control for trailer brakes.
d. Prime mover protection valve with dash control and automatic breakaway feature.
e. Trailer stoplight control operable with foot brake and with hand control for trailer brakes.
f. Two SAE J844 coiled air hoses, not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment.
g. Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to prevent interference with trailer.
h. Dummy gladhand couplers with chains.
i. Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.3 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brakes are furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full-width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. When specified (see 6.2), a tilt type cab with tilting and locking mechanism shall

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be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation, Title 49, Part 399, including the first step height, which shall be not more than 24 inches. Drip rails shall be installed above the cab doors. Safety grips shall be provided on each side of the cab to assist personnel in entering and leaving the cab. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished. For tilt type cabs, provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield.

3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full-width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable driver's seat and an individual passenger seat shall be provided. When specified (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be provided. Interior lighting shall be provided. Two pairs of seat belts shall be installed.

3.4.12.2 Crew cab. When specified (see 6.2), a four-door full width crew cab shall be furnished in lieu of the standard full width cab. The cab shall be equipped with two upholstered, full width seats and backs. The front seat shall be adjustable. Three pairs of seat belts shall be installed for both front and rear seats. Cab doors shall have windows with crank-operated window regulators. Safety grips or grab handles shall be provided on each side of the cab to assist personnel in entering and leaving the cab. Interior lighting shall be provided. The cab roof shall be of one piece construction or, if welded, the roof shall give the appearance of one piece with weld seams being continuous, waterproof, and free of visible seams, bumps or protrusions. Full length drip moldings shall be mounted above doors. When a crew cab is furnished, the cab-to-axle or cab-to-trunnion dimension shall be measured from the back of the crew cab.

3.4.12.2.1 Crew cab seat spacing. With the front seat adjusted to the extreme forward position, there shall be not less than 30 inches, measured in a horizontal plane, between the front of the rear seat back and the rear of the front seat-back. A kickspace height of not less than 2.75 inches shall be maintained between the floor and the rear of the front seat in all positions of adjustment. Leg room and space forward of the front seat shall be equivalent to that provided ahead of the seat in a two-door standard cab.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by air or electric motor(s).

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3.4.15 Front bumper. Unless the front bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided except when a front mounted winch is furnished.

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation, Title 49, section 393.86. A rear bumper shall be provided as specified herein.

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space may be furnished inside the cab. When stowage space for these tools is located outside of the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under an axle or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted. All instruments shall be visible to the driver when seated in the driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Key operated ignition switch.
b. Ammeter, charging indicator or voltmeter.
c. Fuel gage.
d. Oil pressure gage or red indicator warning light.

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e. Engine coolant temperature gage or high coolant temperature and low coolant level red indicator warning lights.
f. High coolant temperature or low coolant level alarm buzzer.
g. Speedometer with recording odometer.
h. Ash receptacle.
i. Dual sunvisors.
j. Front door mounted armrest on driver's side.
k. Driver's compartment ventilator other than window.
l. Tachometer.

3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. The mirrors shall have not less than two supporting arms.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air-operated horn shall be furnished.

3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 Back-up alarm. When specified (see 6.2), the vehicle shall be provided with an audible, pulsating, signaling device (electric or mechanical) to caution personnel when the vehicle is in reverse gear operation.

3.4.24 Front mounted winch. When specified (see 6.2), the vehicle shall be furnished with a winch assembly, mounted on the front of the vehicle and powered by the manufacturer's standard power takeoff. The winch shall conform to SAE J706. The minimum safety factor shall be 2.0 based on the breaking point of the winch. The winch shall be single drum type and shall have one forward and one reverse speed. All winch controls shall be mounted inside the cab. Winch controls shall be located to provide no interference with entry or exit of the driver. Winch shall have a rated single line pull capacity of not less than 20 000 pounds pull on the bare drum. The winch line speed on the bare drum shall be between 15 and 30 feet per minute at an engine speed equal to 35 percent of the engine governed speed. The winch shall be wound with not less than 185 feet of 0.63 inch diameter,

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preformed 6x37, improved plow steel, independent wire rope core (IWRC) cable and shall be equipped with an end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a roller guide. Angles of the winch drive line U-joints shall be not greater than 16.5 degrees.

3.4.24.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 0.25 inch vertical side plates, conforming to the outside radius of the drum flanges. Six bars, 0.38 inch by 1.25 inches, shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the drum flanges shall be not greater than one-half the specified cable diameter.

3.4.24.2 Front bumper. When a front mounted winch is specified (see 3.4.24), the bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either a channel (see 3.4.15) or a pipe type. When a pipe type front bumper is furnished, the nominal diameter shall be not less than three inches, and the minimum wall thickness at any point shall be not more than 12.5% under the nominal wall thickness specified. The pipe type front bumper shall have half-round ball ends.

3.4.24.3 Combination step plate and gravel guard. A combination step plate and gravel guard shall be furnished. The step plate shall be secured to the front bumper and shall be not less than 0.25 inch and not greater than 0.38 inch from the chassis sheet metal. The step plate shall be capable of supporting 300 pounds per square foot. The step plate shall not deflect more than 0.13 inch under the loads imposed.

3.4.25 Air conditioning. When specified (see 6.2), the vehicle shall be equipped with the chassis manufacturer’s standard air conditioning system. The air conditioning system shall include a tinted windshield and tinted glass.

3.4.26 Common locks and keys. When specified (see 6.2), the same key that fits the cab door locks shall fit the starting (ignition) lock. This same key shall fit the door locks and starting locks on all vehicles under contract. Two such interchangeable keys shall be furnished with each vehicle.

3.4.27 Master light shut off switch. When specified (see 6.2), a single switch, in addition to and not in place of the normal light switches, shall extinguish all vehicle exterior lights and the cab interior light, as well as any and all body lights.

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3.4.28 Water cask. When specified (see 6.2), a five gallon insulated, potable water cask shall be furnished. The outside shell shall be galvanized steel or equivalent. The throat size shall be of sufficient area to allow for cleaning the interior of the water cask. The cask shall have a brass, chrome plated, or nylon spigot, flush mounted to prevent damage. Lugs and mounting brackets shall be provided to securely mount the cask in an accessible, vertical position on the vehicle, from which it can be readily removed.

3.4.29 Special equipment. When specified (see 6.2), the following special equipment shall be furnished:

a. Warning devices, highway, triangular, reflective, set of three, with container, conforming to A-A-2128.
b. Utility chain, steel alloy, single leg, minimum 0.75 inch nominal diameter links, not less than 15 feet long with one full capacity grab hook and two full capacity coupling links, all with sufficient capacity for vehicle recovery.
c. Fire extinguisher, 2-3/4 pound capacity, mounted in the rear of the cab on the driver's side, conforming to Federal Motor Carrier Safety Regulation, Title 49, section 393.95.
d. Tire pressure gage, self-contained, 10 to 120 pounds per square inch (psi), suitable for checking tire inflation pressure.
e. Fitting on the air brake system reservoir to accommodate a quick disconnect air hose fitting.
f. Pneumatic hose, tire inflator, not less than 50 feet long, with quick disconnect coupling matching (e) above and necessary fittings for inflating tires.
g. Electric jumper cables, not less than 20 feet long, of not less than 2 gage American Wire Gage (AWG), for jump starting.
h. Tire chains for the outside tires on the rear dual wheels.
i. First aid kit, general purpose, 12 unit, conforming to Drawing 11677011.

3.5 Body. The vehicle shall be equipped with a tank body for transporting and dispensing potable water. The body shall consist of a single compartment tank covered with an outer jacket. The space between the tank and the jacket shall contain insulating material. The tank and all associated fittings, components, and piping shall be free of all dirt, dust, loose scale, and water soluble coating.

3.5.1 Body materials. The tank inner shell, inner heads, internal baffles, internal bulkheads, internal pads, access manholes, and outlets shall be fabricated of type 304L stainless steel conforming to ASTM A240. The tank outer jacket, outer heads, outer pads, catwalks, and all fabricated outer sheet metal shall be made of corrosion-resistant material. All parts, fittings, the pump, and piping in contact with the water shall be made of corrosion-resistant material not affecting the potability of water. Fabrication of the tank and piping system shall be such that it shall not cause or result in contamination of drinking water.

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Rubber or plastic materials used on the interior surfaces of gaskets or hoses that contact potable water shall meet the requirements of the Code of Federal Regulations, Title 21, Part 177, except materials constructed of acrylonitrile and co-polymers shall not be used. Dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action. Teflon packing and tape shall not be used on adjacent parts that are of aluminum content.

3.5.2 Tank. The length of the tank shall be such as to give load distribution proportional to the gross axle weight ratings (GAWRs) of the front and rear axles of the chassis supplied, based on a product density of 8.3 pounds per gallon. Actual capacity of the tank shall provide a minimum of three percent greater volume than the nominal capacity specified in table I to allow for expansion. The tank shell, heads, bulkhead, and baffles shall conform to the requirements of the Code of Federal Regulations, Title 49, sections 178.345 and 178.346 for material, structural integrity, joints, supports and anchoring, circumferential reinforcement, and minimum thickness. Minimum thickness requirements shall be based on a product density of 8.3 pounds per gallon. The tank shall be watertight and shall withstand 3 psi air pressure without leakage. The tank shall be equipped with marked drain plugs located to assure complete removal of water (see 4.1.1).

3.5.3 Baffles. Where the provisions of the Code of Federal Regulations, Title 49, section 178.345 would require circumferential reinforcements, baffles shall be used in lieu of ring stiffeners. Baffles shall be flanged and dished. A half round or full round opening of not less than 3 inch radius shall be provided at the bottom of the baffles. Openings of equivalent size shall be provided at the top of the baffles. Each baffle shall also be provided with not less than an 18 inch diameter manway opening. The edges of the manway openings shall have a flange of approximately one inch. The cross sectional area of each baffle shall be not less than 80 percent of the cross-sectional area of the tank.

3.5.4 Flanging and dishing. The heads and baffles shall be uniformly flanged and uniformly dished.

3.5.5 Tank support. Full length box type, V type or Z type longitudinal subframe members shall be furnished. Full length welding pads shall extend not less than one inch beyond the flanges on the longitudinal subframe members and shall be continuously welded to the tank proper. A flange not less than one inch wide shall be provided at the top of both the outer and inner legs. Flanges shall be continuously welded to the pad the full length of the tank. The width of the upper Z member shall be not less than two inches and shall be continuously welded to the pad the full length of the tank along both sides of the Z. Bolsters shall be properly integrated with the Z type longitudinal and continuously welded to the pads at the front, rear, at each bulkhead and at each baffle.

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3.5.6 Outer jacket and insulation. The tank shall be encircled with not less than 1.5 inch thick insulation or foam type insulation with a K factor of not less than 0.15 and a density of not less than 2.0 pounds per cubic foot. The tank insulation shall be covered with an outer jacket of corrosion-resistant material. The outer jacket does not need to be welded. Overlapping skin sheets can be used with waterproof caulking between sheets, tension fitted using a Pittsburgh Lock Seam.

3.5.7 Manhole and cover. At the top of the tank, a manhole and double cover with an opening not less than 19 inches in diameter shall be provided. The cover shall be removable and mounted on hinges, or shall be bolted. When the cover is bolted, means shall be provided to ensure that it remains attached to the tank. The inner cover shall have an adjustable compression locking clamp; the outer cover shall be insulated and shall include provision for attaching a padlock.

3.5.8 Filler opening. A filler opening of not less than 4 inch diameter tubing shall be provided at the top of the tank, with a removable sleeve strainer that will permit flow of water at the rate of 150 gallons per minute (gal/min). The strainer shall be protected from damage by surge oscillations of water within the tank. The opening shall be provided with a hinged insulated cover. The cover shall be equipped with gaskets and locking clamp to give a leakproof closure and shall have provision for attaching a padlock. All metal parts of the cover exposed to contact with water in the tank shall be fabricated of corrosion-resistant material not affecting the potability of the water.

3.5.9 Vent. A vent having sufficient venting area to allow pumping at the rate of not less than 50 gal/min shall be provided. Vents shall not be located in the filler opening. A permanent, durable plate shall be provided at tank outlets and filler opening, warning the operator to verify that the vents are operating during loading and unloading operations. The vents shall be adequate to limit the tank pressure and vacuum under these conditions to those permitted by the Code of Federal Regulations, Title 49, section 178.346. The vents shall remain sealed when not fulfilling their purpose of equalizing internal and external pressures.

3.5.10 Catwalk. A flat catwalk made of corrosion-resistant material, not less than 24 inches wide and running the full length of the tank body, shall be provided on the top of the tank body. The catwalk shall be covered with a special abrasive type of paint (ship's grip); or shall be of grip-strut material; or shall be open grating type. The catwalk shall be provided with handholds for personnel. Except for the open grating type catwalk, front and rear drains and drain lines on opposite sides of the vehicle and in such other locations as are necessary to prevent liquids from standing on the catwalk or on top of the tank shall be provided. Catwalks, flashing and drains shall prevent the possibility of non-potable water entering the tank. Drain lines shall be internal or external to the tank and shall be of not less than 1-1/2 inch corrosion-resistant tubing.

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3.5.10.1 Access ladder. An access ladder or equivalent access steps, grabhandles and grabrails shall be furnished, providing easy access to and from the catwalk. Ladders, steps, handles and rails shall be functionally spaced and arranged for maximum safety of personnel climbing up onto and down from the catwalk. All components shall be made of corrosion-resistant material.

3.5.11 Emergency valve. A flanged type emergency valve shall be provided and shall be attached to a flanged outlet welded to the bottom of the tank. The method of attachment shall permit complete removal of the valve body. A control with a stop to hold the emergency valve in the open position shall be furnished. The control shall be located in the equipment cabinet. The emergency valve shall have a screen of the smallest mesh that will not restrict the flow of water through the 2 inch discharge piping. The screen shall be removable for cleaning and shall be located inside the tank above the valve mechanism.

3.5.12 Gage. A water level gage shall be provided at the rear near the curbside of the vehicle. The gage shall be calibrated in increments of 50 gallons and shall indicate the actual volume of water in the tank within 50 gallons. When a sight gage is used, it shall be of transparent plastic type material and shall be protected against breakage. Valves shall be provided to seal off the gage in the event of gage failure or malfunction.

3.5.13 Piping and couplings. All piping shall have an inside diameter of not less than two inches. The outlet (emergency) valve from the tank shall discharge into a line leading to the equipment cabinet, arranged with connections and 2 inch gate valves to permit gravity and pressure discharge. At the pump, the piping shall allow use of the pump for filling and for discharging the tank. Piping shall be insulated or protected within the insulated space. The pipe outlet and suction connections (for hose couplings) shall be provided with camlock couplers and chained-on dust covers. Any section of the discharge system, including the pump, which is not self-draining shall be equipped with marked drain plugs. Piping shall not affect the potability of the water. Any couplings between stainless steel piping and galvanized steel piping shall be of the flange type separated by suitable gasketing material. Piping shall be isolated from direct contact with dissimilar metals at the point of connection with the tank body or emergency valve. Piping shall be installed to withstand expansion and vibration without failure and shall conform to the strength requirements of the Code of Federal Regulations, Title 49, section 178.345.

3.5.13.1 Strength of piping and couplings. Strength of piping, fittings, hose and hose couplings for tanks to be unloaded by pressure shall be designed for a bursting pressure of at least 100 pounds per square inch gage (psig) and not less than 4 times the pressure to which, in any instance, that these components may be subjected in service by the action of any vehicle mounted pump or other device (not including safety relief valves). The action of which may be to subject certain portions of the tank piping and hose to pressures greater than the design pressure of the tank. Any coupling used on hose to make connections shall be

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designed for working pressure not less than 20 percent in excess of the design pressure of the hose and shall be so designed that there will be no leakage when connected. Provision for expansion and vibration shall be made in every case to allow for and prevent damage due to expansion, contraction, jarring, and vibration of all pipe. Slip joints shall not be used for this purpose.

3.5.14 Grooved-end pipe clamp couplings or flanges. Clamp couplings or flanges shall conform to A-A-52598 and shall be provided on the inlet and outlet sides of the pump and strainer.

3.5.15 Pipe grooves for couplings. Pipe grooves for couplings shall be in accordance with the dimensions as specified in A-A-52592, or the coupling manufacturer. The groove and outside diameters shall be concentric. The groove and the ends of the pipe shall be deburred to 0.003 to 0.020 inch chamfer or radius. The outside diameter of the pipe between the groove and the end of the pipe shall be smooth without evidence of scale, seams, weld splatter, digs or scratches.

3.5.16 Equipment cabinets. Underslung cabinets shall be provided on both sides of the vehicle, extending from the front end of the tank to the rear wheelhousing.

3.5.16.1 Construction and housing. The cabinet shall house (on the right side of the vehicle) the pipe outlet and suction connections, valves, and controls. The cabinets shall be weatherproof and shall include provisions for flushing and draining. Doors shall be side hung and equipped with hinges. A substantial, theftproof, key-operated, flush type locking handle, with duplicate keys, shall be provided on the doors, with devices to retain doors in the open position. The cabinet shall include a rack or shelves for three 25-foot lengths of 1-1/2 inch discharge hose, and shall be arranged to stow couplings, nozzle, wrenches, and strainer. If the cabinets are attached to the tank shell or heads, continuously welded, corrosion-resistant pads shall be provided at all points of attachment.

3.5.17 Hose tubes. Hose tubes for six 10-foot lengths of 2 inch water suction hose shall be installed along each side of the tank. The hose tubes shall be constructed of corrosion-resistant material not affecting the potability of the water. These tubes may be incorporated in the tank subframe structure. The interior of the tubes shall be free of sharp points or edges. The hose tubes shall be totally enclosed, dustproof, and shall be equipped with hinged access doors with provisions for padlocking.

3.5.18 Pump. A positive displacement rotary pump shall be installed in the equipment cabinet or mounted under the tank and provided with jacketed insulation. When the pump is mounted outside the equipment cabinet, all piping outside the equipment cabinet and leading to and from the pump shall be insulated and jacketed. The parts of the pump in contact with water shall conform to 3.5.1. The pump shall have a capacity of not less than

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50 gal/min against a 100-foot head when pump input shaft is operating at approximately 600 revolutions per minute (rpm). The pump shall be driven from a power takeoff attached to the side of the truck transmission and controlled from inside the cab. The pump shall have two-inch pipe connections for suction and discharge. The suction line shall have a strainer with an opening of not less than 35 percent of its area. A pressure regulating valve shall be provided to bypass the water when the maximum pressure has been reached. An engine speed control device operable from the cab shall be furnished to control the speed of the water pump. The engine speed control device shall be provided with a permanent setting for limiting engine rpm to the speed required for the rated flow capacity of the pump (see 4.1.5).

3.5.19 Equipment and accessories. The following equipment shall be furnished:

a. Six, 10-foot lengths of 2 inch water suction hose. The hose shall be noncollapsible, rubber lined, with a jacket of woven yarn and interwoven round spring wire, covered with a black rubber cover, and shall not contaminate or affect the smell or taste of the potable water. The hose shall be complete with camlock quick couplers as required to assemble all suction hoses to each other, to the pump suction intake, and to the foot strainer. The water suction hose shall be rated for use with potable water. The camlock quick couplers shall be of noncorrosive material.
b. One-foot strainer with check valve for 2 inch hose.
c. Three, 25 foot lengths of 1-1/2 inch discharge water hose. The hose shall be complete with camlock quick couplers as required to assemble all discharge hoses to each other, to the pump discharge, and to the nozzle. The hose shall not contaminate or affect the smell or taste of the potable water.
d. A delivery nozzle or a quick opening-closing faucet with a 1-1/2 inch diameter rigid spout. The nozzle, if furnished, shall be trigger action operated and shall be equipped with a 1-1/2 inch diameter rigid spout. A dust cap complete with security chain shall be furnished for the nozzle or faucet spout.

3.5.20 Body mounting. Unless otherwise specified (see 6.2), the body shall be secured with U-bolts, twin studs, or by brackets, and shall include a breaker strip. All but the rearmost pair of U-bolts, twin studs or brackets shall be spring loaded to permit vertical movement between the chassis and body.

3.5.20.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than three U-bolts or twin studs per side for 4x2 and 4x4 model vehicles and not less than four per side for 6x4 model vehicles, each having a 0.56 inch diameter with a 0.38 inch minimum thread diameter. Tie-plates shall be at least 0.5 inch thick. The vehicle chassis frame shall be braced, using blocks at each mounting point, unless the mounting point is

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located at a full-depth frame crossmember. Blocks shall incorporate a keeper strap or groove for mounting bolt, and shall be of a width and thickness to assure retention. Two tie-back straps shall be provided, one bolted to each side of the rear portion of the body subframe, to maintain body alignment on the vehicle chassis. Forward body mounting bolts shall be located to the rear of the tapered portion of the breaker strips.

3.5.20.2 Brackets. When brackets are used, they shall be bolted to the web of the chassis frame rails. The body mounting brackets shall provide a means of drawing down the body on the chassis rails. Provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure the mounting brackets to the chassis frame rails, the holes shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builder's layouts. Attachments shall neither interfere with nor obstruct existing chassis components.

3.5.21 Alternative body mounting. When specified (see 6.2), the body shall be secured using pre-loaded spring mounting assemblies at the front of the tank, one on each side. Each spring-loaded mounting assembly shall include a mounting bracket incorporating provisions to limit movement of the water tank body during off-road use. A steel-backed rubber isolator pad shall be installed between the bottom of the water tank longitudinal and top of the truck chassis frame rail. The isolator pad shall be a minimum of 30 inches long. A hardwood filler shall be used from the rear of the isolator pad to the end of the water tank body. The isolator pad and hardwood filler shall be minimum of 1 inch thick. The midship and rearmost mounting assemblies are to be twin stud assemblies using 0.75 inch diameter bolts with a 0.5 inch tie plate across the bottom of the truck chassis frame rail.

3.5.22 Rear bumper. A rear bumper fabricated of structural steel rectangular tubing shall be provided. The bumper shall be attached to the truck chassis framerails. The rear bumper should be designed in accordance with Federal Motor Carrier Safety Regulation, Title 49, section 393.86.

3.5.23 Welding. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alignment of lights; adjustment of the engine and brake systems; filling and charging of batteries; alignment of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

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3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted and riveted construction utilized shall be in accordance with the highest standards of the metal fabrication industry. The entire vehicle shall be designed and constructed in accordance with the best known and available practices. Surfaces shall be free of visible grinder marks, rough cut edges, hand hammered edges, and dents. The body shall be free of body filler materials. The tank shell shall be free of ripples except in those immediate weld areas where the heads and baffles are attached to the shell and where the outer jacket pieces are joined together. Permissible ripples in these areas shall be limited to the following:

a. Ripples shall deviate from the surrounding true shape of the shell or jacket not more than 0.13 inch, protruding outward from the true shell or jacket shape or depressed inward from the true shell or jacket shape, but not both immediately adjacent to each other on the same tank shell.
b. Ripples shall transition smoothly and gradually into the surrounding true shape of the shell or jacket.
c. All ripples shall be confined to an area included within a 6 inch band, centered over the weld area and encircling the tank.
d. In addition to the above, ripples shall be confined to those clearly unavoidable by use of first quality manufacturing techniques, careful handling of materials, closely held tolerances and pride of workmanship.

4. VERIFICATION

4.1 First production vehicle inspection. The first vehicle of each model, size and GVW produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.1.1 Joint specimens. To determine conformance to 3.5.2, the contractor shall present for Government inspection and approval joint test specimens prepared from materials representative of those to be used in tanks subject to this specification and by the same technique of fabrication. Two test specimens shall be tested to failure in tension. One pair of test specimens may represent all the tanks to be made of the same combination of materials by the same technique of fabrication, and in the same shop, within 6 months after the tests on such samples have been completed. The butt welded specimens tested shall be considered qualifying other types or combinations of types of weld using the same filler material and welding process as long as parent metals are of the same types of material.

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4.1.2 Vehicle weight. To determine conformance to 3.2.4 through 3.2.6, the first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on front and rear axles. The total imposed loading on front and rear axles shall be determined by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds and the payload required to provide the specified GVW. The determined imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

4.1.3 Road test. To determine conformance to 3.3 through 3.3.2, 3.4.1.5, 3.4.9.3, and 3.4.11, the vehicle, for models built to 4-wheel drive while on the chassis manufacturer's production line, shall be road tested by the contractor with and without payload. The road test shall include at least 25 miles of operation on gravel or dirt roads, half with the vehicle loaded and half with the vehicle empty. The vehicle, for 4-wheel drive models qualifying under 3.1.1.12, also shall be road tested with and without payload. The road test for conversion units for each of the two conditions (loaded and unloaded) shall be for not less than 30 miles at speeds up to 50 mph over highways and gravel roads and for not less than 5 miles at speeds up to 15 mph over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop no less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of vehicle components during the road test shall be cause for rejection. Failure includes permanent deformation as well as breakage.

4.1.4 Leakage test. The completely assembled tank body, including all piping and hoses, shall be tested for leaks at design pressure in accordance with DoT Safety Regulation, Title 49, section 178.346 after the road test. In addition, all piping and hoses shall be tested for leaks at the maximum pressure they can encounter during use of the vehicle mounted pump.

4.1.5 Pump test. The pumping equipment shall be tested with potable water to determine conformance to the requirements specified in 3.5.18.

4.1.6 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

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4.2 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken. Following corrective action, the vehicle shall be reinspected and any previously performed tests, at the Government's option, shall be repeated.

4.3 Inspection of production vehicles. All vehicles under contract shall be tested for leaks as specified in 4.1.4. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The vehicles covered by this specification are intended for nontactical use by the Government for hauling and dispensing potable water.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Model, size and GVW of vehicle required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. Exterior color and identification markings for the appropriate military service (see 3.1.1.1).
e. Concealed markings, if required (see 3.1.1.1).
f. Rustproofing, if required (see 3.1.1.2).

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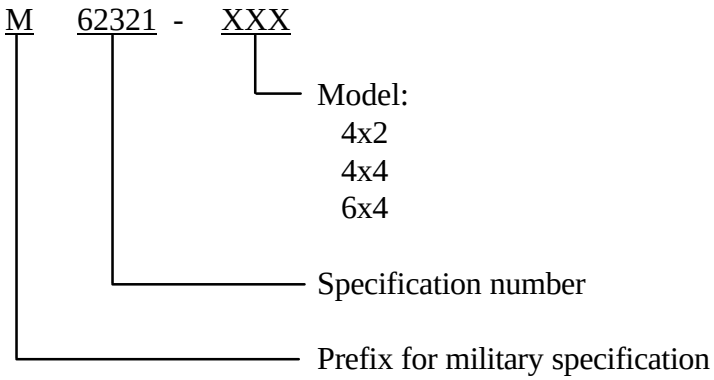
g. Type of tropical rustproofing, if required (see 3.1.1.2).
h. Drive hubs, if required for 4x4 model vehicles (see 3.1.1.5).
i. Trailer towing package, if required (see 3.1.1.7).
j. A seven-pin, trailer lighting cable, if required (see 3.1.1.8.).
k. Satisfactory operation on JP-8 fuel, if required (see 3.4.1.1).
l. Gasoline engine, if required (see 3.4.1.2).
m. Silicone rubber radiator and heater hoses, if required (see 3.4.1.8).
n. Power plant heaters and fuel warmer, if required (see 3.4.1.9).
o. If a battery heater is not required (see 3.4.1.9).
p. Auxiliary 24-volt system with trailer receptacle, if required (see 3.4.2.6).
q. Automatic or semiautomatic transmission, if required (see 3.4.5.3).
r. Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
s. Two-speed axle, if required for 4x2 or 6x4 model vehicles (see 3.4.9.1).
t. Traction control, if required (see 3.4.9.3).
u. Wide base type tires and wheels, if required (see 3.4.10).
v. Cast spoke type wheels, if required (see 3.4.10).
w. Tire size, load, and percentage of usage for procurement or replacement tires (see 3.4.10.1).
x. A carrier for spare tire assembly, if required (see 3.4.10.2).
y. A spare wheel or rim, if required (see 3.4.10.3).
z. Spare tire assembly, if required (see 3.4.10.4).
aa. A trailer brake control system, if required (see 3.4.11.2).
ab. Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
ac. Tilting hood and fender assembly, if required (see 3.4.12).
ad. Tilt type cab, if required (see 3.4.12).
ae. Individual adjustable driver and passenger seats, if required (see 3.4.12.1).
af. Air ride driver's seat, if required (see 3.4.12.1).
ag. Crew cab, if required (see 3.4.12.2).
ah. Tools, if required (see 3.4.16.1).
ai. Air-operated horn, in addition, if required (see 3.4.21).
aj. Back-up alarm, if required (see 3.4.23).
ak. A front mounted winch, if required (see 3.4.24).
al. Air conditioning system, if required (see 3.4.25).
am. If cab door key should fit starting (ignition) lock (see 3.4.26).
an. Master light shut off switch, if required (see 3.4.27).
ao. Water cask, if required (see 3.4.28).
ap. If special equipment should be furnished (see 3.4.29).
ar. If body mounting should be other than as specified (see 3.5.20 and 3.5.21).
as. Packaging requirements (see 5.1).

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6.3 Alternate fuel. Grade JP-8 fuel is in accordance with MIL-T-83133.

6.4 Classification change. Truck, tank, potable water, 2200 gallon, diesel engine driven, 34 500 pounds GVW, 6x4 of MIL-T-62322A, dated 12 July 1982, may now be procured under model 6x4, size 2200, 34 500 pounds GVW of this specification.

6.5 Part or Identifying Number (PIN). The PINs to be used for trucks acquired by this specification are created as follows:



6.6 Subject term (key word) listing.

Catwalk
Dispensing
Four-wheel drive
Front mounted winch
Nontactical
Rear-wheel drive
Rotary pump

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - MC
Air Force - 99

Preparing Activity:
Army - AT

(Project 2320-0707)

Review Activity:
Air Force - 84

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-62321D	**2. DOCUMENT DATE *(YYMMDD)*** 970320

3. DOCUMENT TITLE Trucks, Tank: Potable Water, 1000 and 2200 Gallon, Diesel and Gasoline Engine Driven, 19 000 and 34 500 Pounds GVW, 4x2, 4x4, and 6x4, Commercial

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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